

SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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ABSTRACT

Self-evolving language-model agents increasingly treat skill identities as control objects: a sampled skill can determine the next training task, a fresh ID can receive retrieval priority, and feedback can be written back to the selected package. We ask whether such control should change when semantic capabilities are merely split or cloned across package identities. We formulate *Skill-Taxonomy Representation Invariance* (STRI). For exact refinements, we prove that invariance is equivalent to factoring aggregate semantic-class mass through the quotient; any strictly positive identity-local score followed by normalization is therefore clone-sensitive. Released Skill-SP instantiates this failure mode at initialization: 15 packages receive probability $1/15$ each, double-supported Level-1 rows receive twice the additive eligibility opportunity of singleton rows, and a counterfactual same-state sampler-input clone changes its prompt-mixture by $TV = 0.0583$; quotient-conserved mass restores it exactly. For partial overlap, define $R^*(A; q) = \min\{t : q \leq Aw \leq tq, w, t \geq 0\}$ for support matrix A and representation-independent target $q > 0$. It equals one iff q lies in the package support cone. Skill-SP Level-1 has neutral $R^* = 2$ on full, calibration, and tool-disjoint heldout matrices; exact edit-radius, rewiring, target-ray, and witness-peeling controls preserve the diagnosis under their stated assumptions. Level-3 and the 127/128-overlap logical domain are equalizable. Across these positive and negative regimes, target non-realizability in the package support cone—not overlap prevalence—is the audited discriminator. In one archived AutoSkill P19 substrate, splitting one skill into four exact-semantic IDs deterministically crowds out a post-checkout skill and removes a mechanically defined destructive behavior (6/6 original versus 0/6 split), while ID-placebo and quotient controls restore it (3/3 each; Fisher $p = 0.00108$). Under the split representation, adding that post-checkout skill back restores the behavior in 3/3 fresh runs while a matched cleanup add-back yields 0/3 (exact one-sided Fisher $= 1/20$), isolating the mediator within this substrate. Thus representation can alter executed behavior on this bounded substrate without semantic change. A semantic-first construction marks the action-basis escape when package-only control cannot realize q ; it is constructive, not a validated repair.

1 INTRODUCTION

Many self-evolving agents adapt through external state—memories, skills, workflows, tools, prompts, retrieval policies, or harness code—rather than only through foundation-model weights. Skills are especially attractive because a skill can package reusable procedure, metadata, examples, applicability information, and verification logic. Recent systems make the skill identity itself a control variable: Skill-SP (Huang et al., 2026) samples a skill before generating a training task, while SkillRL (Xia et al., 2026) gives dynamically created skill IDs priority in a fixed retrieval budget.

This architecture makes the *representation* of the skill library consequential. Consider two taxonomies that expose the same executable semantic capability. One stores a primitive once; another stores an

exact clone under a fresh ID. One represents generic and specific capabilities separately; another groups unchanged primitives behind a macro package. If no new primitive behavior or support region is added, should the future curriculum, retrieval exposure, or credit allocation change solely because package bookkeeping changed?

Prior work establishes that skill availability, representation, retrieval, composition, and exposure matter (Xu & Wu, 2026; Jiang et al., 2026; Gao, 2026; Singh & Joachims, 2018; Biega et al., 2018); Sec. 2 separates our support-fixed estimand from those objects. The challenge is that overlap and skill count themselves change under refactoring. We therefore work asset-first: freeze independent support truth, quotient exact-clone nuisance, overgrant the strongest same-information package baseline, and ask whether the remaining support geometry can realize a representation-independent target.

Released Skill-SP. On API-Bank Level-1, 314 released tasks are covered by at least one released validator and 183 have multiple memberships. The release’s name+description Jaccard duplicate filter at threshold 0.33 misses all five observed specific–generic support-overlap pairs. Minimum-cardinality whole-package pruning can reduce the active library from 10 to 7 while preserving exact support, but 71 overlapping tasks remain. The residual is therefore not merely an exact-clone artifact.

Released SkillRL. In released template-mode retrieval, dynamic general skills are included before the remaining fixed $\text{top}_k=6$ budget is filled with static skills. Adding one exact-content fresh-ID copy at a time admits all 12 tested clones. Across 223 released ALFWorld task descriptions, 11/12 clone targets change the unique semantic retrieval set and 5/12 reduce the number of unique general-skill contents returned.

These systems establish representation sensitivity, but not its boundary. The logical domain provides a direct counterexample to overlap-as-sufficient-explanation: 127/128 tasks overlap while the neutral target remains exactly realizable. Thus overlap count alone cannot determine the STRI residual.

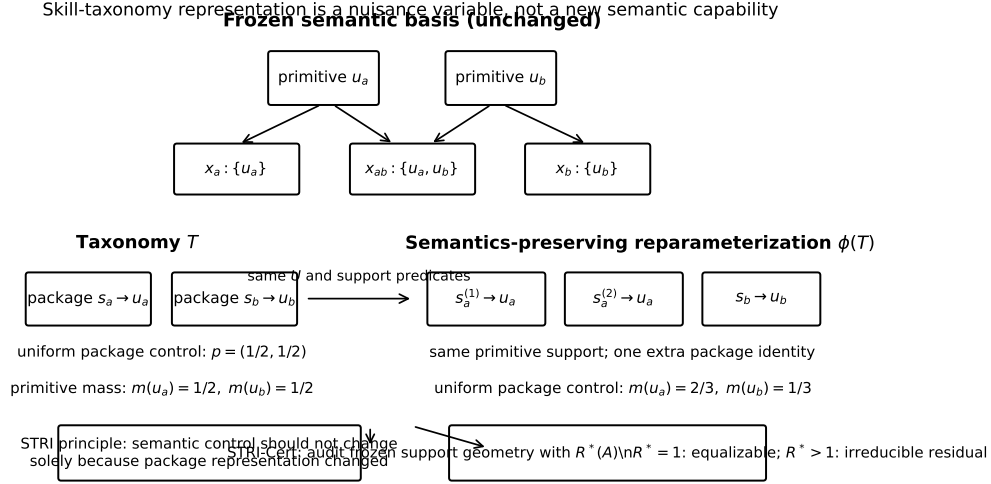
Contributions.

1. **A support-fixed systems invariant.** We ask whether a skill-indexed self-evolution controller changes under package refactorings that preserve semantic support. The novelty claim is this invariant and audit object, not skill representation or exposure allocation in general.
2. **An exact certificate.** Quotient factorization characterizes exact-refinement invariance and $R^*(A; q)$ decides package-first target realizability; semantic-first selection marks the action-basis escape. We reuse classical convex feasibility/duality rather than claim a new LP or cone theorem.
3. **Released-system evidence with explicit boundaries.** Skill-SP/SkillRL expose identity sensitivity; positive and negative support geometries separate overlap from realizability; one frozen AutoSkill P19 intervention traces a bounded representation-to-behavior path. Exact edit radii, degree-preserving rewirings, target rays, and witness peeling quantify robustness without expanding to general utility or safety.

2 RELATION TO EXISTING WORK

Controlled skill representation changes a different variable. SkillsBench varies skill availability and presentation granularity (Xu & Wu, 2026); Demystifying Agent Skills isolates representation, annotation, retrieval difficulty, and cross-framework robustness (Jiang et al., 2026); composition work maps task decompositions to multiple skills (Gao, 2026). They establish that presentation and retrieval matter, but do not hold semantic support fixed while perturbing only package bookkeeping. STRI therefore asks an invariance question, not which representation has the best downstream score.

Self-evolving controllers are the closest substrate. Skill-SP (Huang et al., 2026) is closest: it samples a skill before task generation and releases independent validators. STRI then overgrants its package controller, allowing any nonnegative mass under the full frozen support matrix. SkillRL (Xia et al., 2026) supplies an identity-sensitive fixed-budget retrieval phenotype but no independent A ; AutoSkill (Yang et al., 2026) supplies one bounded behavioral seam. Work on redundancy, failures,



126 Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 6 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

132 routing, and skill/harness co-evolution motivates the setting but not this support-fixed audit (Zhu et al., 2026; Liu et al., 2026; Mao et al., 2026; Dong et al., 2026; Chen et al., 2026; Wang et al., 2026).

135 **Optimization boundary.** Fair-exposure methods allocate attention (Singh & Joachims, 2018; Biega et al., 2018); our covering/cone/duality machinery is classical (Lovász, 1975; Sinkhorn & Knopp, 1967; Radunović & Le Boudec, 2007; Beck & Fiala, 1981; Boyd & Vandenberghe, 2004). We claim the systems invariant, information boundary, and audit mapping—not a new LP, cone theorem, or fairness objective.

142 3 METHOD: FROM PACKAGE IDENTITY TO SEMANTIC EXPOSURE

144 3.1 PROTOCOL AT A GLANCE

146 **Design intuition.** Package j contributes support column $A_{\cdot j}$, so package mass w induces semantic exposure Aw . In Fig. 1, an exact clone is therefore a duplicated support column: it may perturb identity-indexed package mass, but should not create a new semantic degree of freedom. STRI-Cert separates two questions: exact identity refinements should vanish after quotienting clones; afterward, a target q is package-realizable exactly when it lies in the nonnegative cone of the remaining columns.

151 The five-step order isolates those questions: **freeze support/target provenance** $\rightarrow$ **quotient exact clones** $\rightarrow$ **overgrant arbitrary $w \geq 0$ with full frozen $A \rightarrow$ solve primal/dual ($R^* = 1$ constructive weighting; $R^* > 1$ tight residual plus witness) $\rightarrow$ separate diagnosis from repair.** Semantic-first control can escape cone infeasibility only by changing the action basis, so it is a constructive boundary witness rather than validated downstream repair.

157 3.2 FROZEN SUPPORT MATRIX

158 Let $X = \{x_1, \dots, x_n\}$ be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let $S = \{s_1, \dots, s_m\}$ be the current skill packages. Define the binary incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

Support is frozen before STRI: released Skill-SP validators define the tool-call matrix, while released compiler specifications plus fixed CSP/contract checks define the logical matrix. Certificates cover only rows with at least one supported package; zero-support rows are reported separately rather than misclassified as representation distortion.

At a frozen pre-task state, package-only control assigns $w \in \mathbb{R}_{\geq 0}^m$ and induces eligibility exposure $e = Aw$. Exposure is control opportunity, not task probability or utility. The target $q \in \mathbb{R}_{> 0}^n$ is a *representation-independent semantic target*: it cannot change because packages are renamed, cloned, split, or regrouped. Without an externally justified preference we use neutral $q = \mathbf{1}$.

3.3 SAME-INFORMATION PRE-TASK CONTROLLERS REDUCE TO PACKAGE MASS

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state z , every randomized same-information rule—including a bandit, MLP, or neural router—induces a categorical distribution over packages, hence one nonnegative mass vector $w(z)$. The strongest same-information baseline is therefore arbitrary nonnegative package mass with access to the complete frozen support matrix.

This is an intentional overgrant: the baseline may ignore the released scoring rule and choose any $w(z)$ using complete frozen A and the same pre-task information. If this class has $R^*(A; q) > 1$, deduplication, retuning, bandits, or neural package routers cannot remove the residual without new information or a different action basis. The reduction is pointwise in z ; it neither collapses a longitudinal controller to one static vector nor permits conditioning on the future task.

3.4 EXACT-REFINEMENT STRI FACTORS THROUGH A QUOTIENT

Let $\mathcal{R}$ be a package representation and let $Q(\mathcal{R})$ collapse exact semantic clones—packages with identical semantics, support, and controller-relevant state—into classes. Let $G_{\mathcal{R}}$ sum a package-mass vector over these classes. For a fixed pre-context state z , define *exact-refinement STRI* by

$$G_{\mathcal{R}}p_{\mathcal{R}}(z) = G_{\mathcal{R}'}p_{\mathcal{R}'}(z) \quad \text{whenever } Q(\mathcal{R}) = Q(\mathcal{R}').$$

Proposition 1 (quotient-factorization characterization). A controller family satisfies exact-refinement STRI iff there exists a quotient-level map $\bar{p}$ such that $G_{\mathcal{R}}p_{\mathcal{R}}(z) = \bar{p}(Q(\mathcal{R}), z)$ for every representation and state.

Proof. The forward direction defines $\bar{p}$ using any representative of a quotient class; STRI makes this definition representation-independent. The reverse direction is immediate because equal quotients give equal $\bar{p}$. $\square$

Corollary 1 (identity-local normalization is clone-sensitive). Suppose $p_j = g(h_j, z) / \sum_k g(h_k, z)$ with $g > 0$ and an exact clone receives the same local state h_j . Writing its score as $a > 0$ and all other scores as $B > 0$, the semantic-class mass changes from $a/(a + B)$ to $2a/(2a + B)$, an increase of $aB/((a + B)(2a + B)) > 0$. Repeated clones drive class mass toward one, so exact-refinement invariance requires class-aware conservation or another quotient-factorized rule.

Proposition 2 (semantic-first construction handles arbitrary overlap). Normalize a target ray $q > 0$ to $\pi = q/(\mathbf{1}^\top q)$. If every target-positive semantic unit i has at least one supporting package, choose any row-stochastic kernel K with $K_{ij} = 0$ whenever $A_{ij} = 0$, then sample $i \sim \pi$ before sampling implementation $j \sim K_i$. The semantic marginal is exactly π for arbitrary overlap, and any clone/split can redistribute K_i within the corresponding implementation class without changing that marginal.

Proof. The joint mass is $J_{ij} = \pi_i K_{ij}$, so $\sum_j J_{ij} = \pi_i \sum_j K_{ij} = \pi_i$; support-constrained row-stochastic K exists exactly when every positive-target row is covered. $\square$ This changes the action interface: it requires representation-independent semantic intent and conditional implementation. It is a constructive action-basis witness for escaping package-first infeasibility, not a same-information baseline or validated downstream method.

4 TARGET REALIZABILITY AND STRI-CERT

4.1 TARGET-CONDITIONED FINITE-SUPPORT CRITERION

Define the multiplicative distortion from the target ray q to the package support cone by

$$R^*(A; q) = \min_{w \geq 0, t \geq 0} t \quad \text{s.t.} \quad q \leq Aw \leq tq. \quad (1)$$

We write $R^*(A) \equiv R^*(A; \mathbf{1})$ for the neutral audit used in our released-system tables.

Theorem 1 (target realizability in the support cone). For any $q > 0$, $R^*(A; q) = 1$ if and only if $q \in \text{cone}(A) = \{Aw : w \geq 0\}$. Otherwise $R^*(A; q) > 1$ is the tight minimum componentwise multiplicative distortion from q achievable by package-only mass.

Proof. The constraints force $t \geq 1$. At $t = 1$ they reduce exactly to $Aw = q$, which is equivalent to $q \in \text{cone}(A)$. For any feasible w , the smallest admissible t is $\max_i (Aw)_i / q_i$ after scaling w so that $\min_i (Aw)_i / q_i = 1$; minimizing over w gives the tight distortion. $\square$

This separates STRI from a uniformity assumption: $q = \mathbf{1}$ is the neutral target, not the definition of representation invariance. It also makes exact clone/split invariance immediate: duplicating a support column does not change $\text{cone}(A)$, so it cannot change $R^*(A; q)$ for any fixed q . The cone-membership statement itself is classical convex feasibility; STRI contributes the system invariant that makes this feasibility question the relevant audit object for a self-evolving skill controller, not a new theorem about convex cones.

Proposition 3 (dual certificate). The dual of Eq. (1) is

$$D^*(A; q) = \max_{\alpha, \beta \geq 0} q^\top \alpha \quad \text{s.t.} \quad A^\top (\beta - \alpha) \geq 0, \quad q^\top \beta \leq 1. \quad (2)$$

Strong duality gives $D^*(A; q) = R^*(A; q)$. Thus every dual-feasible pair is a certified lower bound, while an optimum is complete even when no local witness template is available. On every audited neutral-target regime the numerical primal–dual gap is below 10^{-8} .

4.2 HUMAN-READABLE FACTOR-2 WITNESS

Theorem 2 (global-singleton overlap lower bound). Suppose packages a, b each have a context supported by exactly that one package over the full frozen support universe, and another context is supported by both a and b , possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore $R^*(A) \geq 2$ for the full support matrix.

Proof. Let $r > 0$ be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly $\{a\}$, its exposure is $w_a \geq r$; likewise $w_b \geq r$. The shared context contains both a and b , and additional packages have nonnegative weight, so its exposure is at least $w_a + w_b \geq 2r$. Thus every w has focal max/min ratio at least two. For the same w , the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over w gives $R^*(A) \geq 2$. $\square$

For the neutral target $q = \mathbf{1}$, Theorem 2 is also a sparse dual certificate: put $\alpha = 1$ on the two singleton rows, $\beta = 1$ on their shared row, and zero elsewhere. Dual feasibility follows from the support pattern and the objective is two. In released Skill-SP Level-1, the LP dual recovers such a sparse optimum and the primal attains $R^* = 2$.

4.3 STRI-CERT AND DEPLOYABLE VARIANTS

STRI-Cert is a diagnostic protocol, not a learned predictor or a new LP solver. Its output has three operational meanings: **realizable** ($R^* = 1$) means package-only retuning is sufficient *in principle* to hit the declared target; **residual** ($R^* > 1$) means no same-information package router can hit

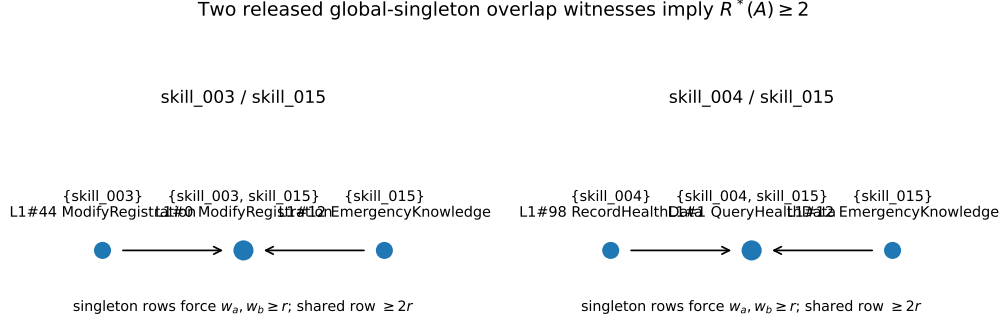


Figure 2: Two released global-singleton overlap witnesses. In each case, retaining positive exposure on the two singleton regions forces at least twice that minimum exposure on the shared region.

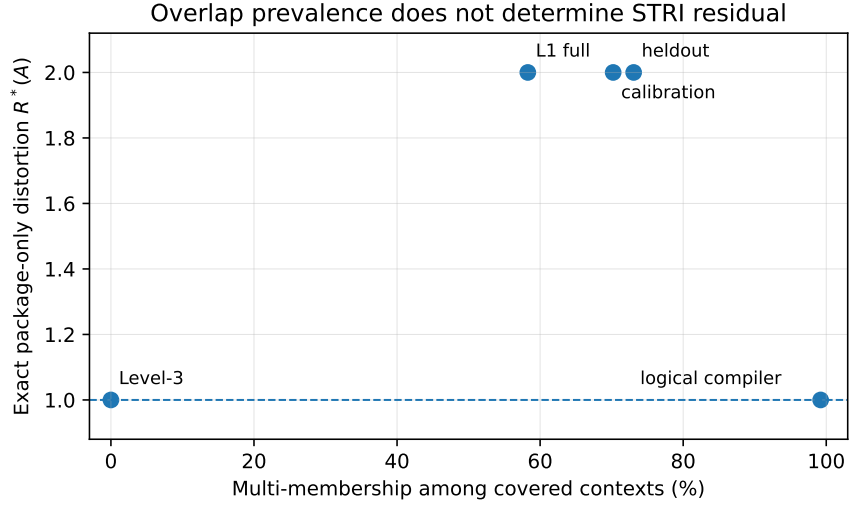


Figure 3: Overlap prevalence does not determine STRI residual. The three Level-1 regimes independently have $R^*(A) = 2$, while Level-3 is equalizable with disjoint support and the logical compiler domain is equalizable despite 127/128 multi-membership tasks.

that target without changing information, assumptions, or the action basis; and **fail-closed** means support/target provenance or optimization is not valid enough to make either statement. The dual may expose a sparse obstruction, while the primal gives a constructive weighting in the realizable case. HiGHS solves both programs with paper-level tolerance 10^{-8} ; audited primal-dual gaps are at most 10^{-8} . The certificate concerns feasibility, not downstream utility or a unique weight vector.

Two variants make deployment assumptions explicit. With independently justified graded support $0 \leq L \leq A \leq U$, the box-robust audit uses $Lw \geq q$ and $Uw \leq tq$; estimating valid bounds remains an identification problem. Adding $w_j \leq \rho \sum_k w_k$ audits package-mass concentration via $R_\rho^*(A; q) \geq R^*(A; q)$. Both remain linear programs.

5 EXPERIMENTAL SETUP

Evidence, estimands, and controls. Predictions are direct: clone/split refactoring leaves the support cone and R^* unchanged; overlap alone need not imply residual; a factor-2 witness rules out all same-information package weights for neutral q . Skill-SP tests positive/transfer regimes; Level-3/logical compiler test negative boundaries.

Table 1: Two-stage audit across released systems. We test observable identity sensitivity first and run STRI-Cert only when independent support truth exists; absence of a support contract is not repaired with candidate-defined labels.

System	Identity perturbation	Independent A ?	Audit role
Skill-SP	Pristine sampler gives $2\times$ eligibility; sampler-input clone changes class mass $1/15 \rightarrow 1/8$, while quotient conservation restores it exactly	Yes: released validators	Controller violation + exact clone repair + geometry certificate
SkillRL	Fresh-ID exact clones: 11/12 change semantic retrieval over 223 tasks	No explicit support contract	Phenotype + qualified downstream bridge; no R^*

Table 2: Matched baseline ladder on the Skill-SP tool-call support matrix. Every baseline uses the same frozen released support information; the ladder deliberately strengthens from name-level duplicate removal to the exact best context-independent package reweighting.

Baseline / reduction	Outcome	Evidence
Released name+description Jac-card duplicate filter	Does not absorb	Misses all 5/5 observed specific-generic support-overlap pairs; similarity 0.1429–0.20 is below the released 0.33 threshold.
Exact semantic-clone deduplication	Does not absorb partial overlap	The positive regime uses distinct specific/generic packages with different unique support regions.
Minimum-cardinality whole-package pruning preserving exact support	Partial reduction only	Active packages drop from 10 to 7, but 71 overlap rows remain while exact support is preserved.
Arbitrary nonnegative global package weights	Exact neutral-target residual	The exact optimum is $R^*(A; \mathbf{1}) = 2$.
Global-singleton overlap witness	Sparse dual lower bound	Two independent released witnesses imply $R^*(A; \mathbf{1}) \geq 2$, and the exact LP attains 2.

Six evidence roles. Structural audits use neutral $R^*(A)$ with primal/dual tolerance 10^{-8} . E1 is the released-system result; E2 quotient/baseline ablations; E3 transfer/negative regimes plus SkillRL’s phenotype-only boundary and AutoSkill’s bounded P19 intervention; E4 support/target sensitivity; E5 witness redundancy and the P19 mediator trace; E6 conditional solver cost. Offline completion uses only frozen support or deterministic synthetic matrices—zero new annotations, task outcomes, model calls, or GPU runs.

6 EXPERIMENTS AND RESULTS

6.1 SKILL-SP TOOL-CALL SUPPORT AND RELEASED-CONTROLLER CONSEQUENCE

API-Bank Level-1 has 314 covered rows, 183 with multiple memberships, and neutral $R^* = 2$. The released pre-task sampler initializes all 15 tool-call packages at probability $1/15$, so a double-supported row receives $2/15$ eligibility versus $1/15$ for a singleton. The source paper reports a -1.9 point accuracy change when dynamic routing is replaced by uniform routing (Huang et al., 2026); we use this only to establish that allocation is operational, not to attribute utility to STRI.

For a same-state sampler-input reparameterization, we copy one active package with identical content, support, statistics, and sampler state under a fresh ID and rerun the author’s weighting function. This is not a claim that the induction path would admit a literal duplicate. The cloned semantic class moves from $1/15$ to $2/16 = 1/8$, yielding prompt-mixture $TV = 7/120 = 0.0583$ because package ID is absent from the questioner message. Quotient-conserved mass instead splits the original $1/15$ class mass and restores prompt TV exactly to zero. These are input-mixture statements, not generated-task or utility claims.

A frozen tool-disjoint test gives the same structural answer: calibration has 47 covered rows/33 multi-memberships and $R^* = 2$; heldout has 52/38 and $R^* = 2$. The two released factor-2 witnesses are

Table 3: Neutral-target ($q = 1$) mechanism boundary across real first-party support regimes. High overlap is neither necessary nor sufficient; the discriminator is target realizability in the support cone.

Regime	Covered	Multi	Overlap	R^*	Certificate
API-Bank Level-1 full	314	183	58.3%	2.0	Residual
Level-1 calibration tools	47	33	70.2%	2.0	Residual
Level-1 tool-disjoint heldout	52	38	73.1%	2.0	Residual
API-Bank Level-3	34	0	0.0%	1.0	Equalizable
Logical compiler validation	128	127	99.2%	1.0	Equalizable

`skill_003/skill_015` (singleton L1#44, L1#12; shared L1#0) and `skill_004/skill_015` (singleton L1#98, L1#12; shared L1#1), matching the exact LP.

6.2 LEVEL-3: DISJOINT-SUPPORT NEGATIVE CONTROL

Level-3 provides a natural released negative regime. Thirty-four rows are covered, every covered row has exactly one skill membership, and $R^* = 1$. Equal weights on the active packages equalize every covered task. STRI-Cert therefore does not mechanically label every released taxonomy problematic.

6.3 LOGICAL COMPILER DOMAIN: HIGH OVERLAP BUT NO RESIDUAL

The first-party logical release gives a stronger negative control: 128/128 programmatically generated units pass the author’s compiler/uniqueness contracts and are cross-checked against all eight released support specifications, with no language model. Although 127/128 units have multiple memberships across 38 support patterns, $R^* = 1$ because `skill_008` supports every unit. This equalizer is concentrated: the max-share audit gives $R_{0.75}^* = 4/3$, $R_{0.5}^* = 2$, and $R_{0.25}^* = 4$. High overlap is therefore not sufficient; realizability depends on support geometry and the declared feasible package-mass set.

6.4 SKILLRL: INDEPENDENT PHENOTYPE, BOUNDED ROLE

SkillRL confirms identity sensitivity in a different control surface: fresh-ID exact-content clones are admitted for 12/12 tested skills, 11/12 change semantic retrieval over 223 ALFWorld descriptions, and 5/12 reduce unique general-skill content under `top_k=6`. It does not expose independent support truth for R^* , so we do not retrofit a candidate-defined A ; a qualified endpoint bridge found no success disagreement beyond placebo and is not a population-level no-effect theorem.

6.5 AUTO-SKILL: DYNAMIC BEHAVIORAL PROPAGATION

AutoSkill (Yang et al., 2026) exposes a first-party `sdk.search` seam. In one frozen five-skill P19 library at `top_k=5`, a four-way exact-semantic split crowds out a post-checkout skill; ID renaming preserves, and exact-semantic quotienting restores, the original semantic set. The mechanical post-checkout behavior occurs in 6/6 original, 0/6 split, 3/3 placebo, and 3/3 quotient runs (one-sided Fisher $p = 0.00108$). Under the split, adding the crowded-out skill back restores behavior in 3/3 fresh runs versus 0/3 for a matched cleanup add-back ($p = 1/20$). This isolates one mediator on one substrate, not AutoSkill utility, safety, or prevalence.

6.6 MATCHED BASELINES, REPRESENTATION ABLATIONS, AND FAILURE ANALYSIS

Table 2 gives the same-information ladder. Minimum exact-coverage pruning removes 61.2% of Level-1 overlap (183 to 71 multi-membership rows), yet arbitrary nonnegative package mass with the full support matrix still has $R^* = 2$ on full, calibration, and heldout regimes. Exact column cloning/splitting leaves the support cone and $R^*(A; q)$ unchanged; quotienting restores the original support rows and interpretable witness layer (Figure 4A).

Across 49 Level-1 tools, 4 have the closed-form witness and all are residual; all 10 overlap-without-witness tools are equalizable by the exact LP. One-cell stress is asymmetric: all 1,387 additions and all 366 non-uncovering deletions keep Level-1 residual, whereas Level-3 turns residual under all 102

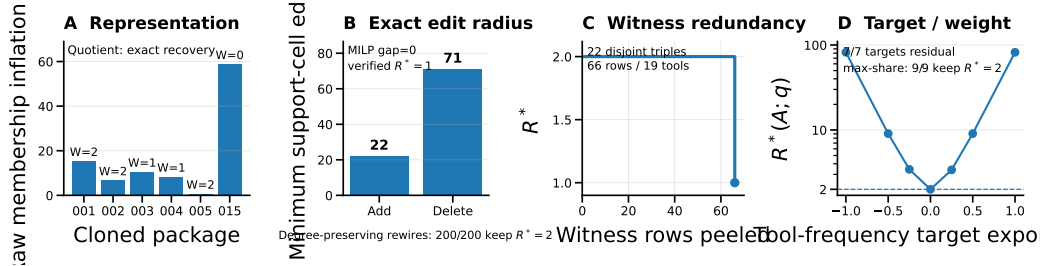


Figure 4: Structural robustness. **A:** clone perturbation and quotient recovery. **B:** exact edit radii are 22 additions/71 deletions; 200/200 degree-preserving rewirings retain $R^* = 2$. **C:** the residual survives 22 disjoint three-row witnesses spanning 19 tools. **D:** 7/7 frozen target rays remain residual and max-share $\rho \in [1/6, 1]$ leaves $R^* = 2$.

additions and the logical domain breaks under 127/595 valid deletions. Earlier leave-one-out and 500 fixed-seed tool bootstraps remain consistent.

E4: structured support, target, and weight sensitivity. All 184 tool-block $\times$ package biases and 49/56 observed-signature relabels remain residual; only 3/9 package-wide coverage-preserving variants do. Exact neutral-target MILPs require at least 22 added cells (the `skill_015` completion) or 71 deletions to reach $R^* = 1$. Stronger controls agree: 200 degree-preserving bipartite rewirings all retain $R^* = 2$; seven frozen representation-independent tool-frequency target rays are all residual (neutral q is least distorted at $R^* = 2$); and max-share $\rho \in [1/6, 1]$ leaves $R^* = 2$. These are sensitivity controls, not a randomization p -value or deployment-priority claim.

E5: sparse-witness redundancy. Iteratively peeling all positive-mass rows of one tight sparse dual optimum leaves $R^* = 2$ through 22 pairwise-disjoint three-row witnesses; after 66 rows spanning 19 tools (21.0% of Level-1) are removed, the remainder reaches $R^* = 1$. This is a solver-path redundancy stress, not minimum deletion or witness uniqueness.

6.7 E6: CONDITIONAL SOLVER COST

On deterministic covered synthetic matrices, dense SciPy/HiGHS grows from 0.00234 s median at 128×8 to 0.685 s at $16,384 \times 96$ (largest max 0.765 s; 32,768 inequalities; 24.25 MiB). This host-load-dependent profile makes no asymptotic or solver-novelty claim.

7 DISCUSSION, LIMITATIONS, AND TAKEAWAY

Evidence separates three claims: Skill-SP/SkillIRL show identity sensitivity; with independent support, Skill-SP/Level-3/logical regimes show realizability geometry—not overlap count—separates residual from equalizable cases; AutoSkill provides one bounded representation-to-behavior existence proof. STRI-Cert is conditional on A, q and triages action rather than automatically correcting taxonomies: $R^* = 1$ keeps package retuning viable, $R^* > 1$ rules out that same-information class for the declared target, and invalid provenance redirects effort to identification. The bounded AutoSkill path and SkillIRL phenotype neither supports population utility, safety, or regret; semantic-first remains constructive, not a validated repair. Takeaway: quotient representation nuisance, audit realizability, then repair the identified assumption or interface.

AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in a preregistered Qwen3-8B dynamic pilot; that pilot failed its frozen author-native qualification gate, remains documented in the supplement, and is excluded from scientific claims about dynamic STRI. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The main paper states the exact-refinement quotient characterization, semantic-first design construction, target-conditioned primal/dual programs, closed-form witness, box-robust extension, and constrained-controller audit used for the load-bearing claims. The Skill-SP controller audit is bound to author commit `bb693c89fee66e1f824d6a777759a49b7a295a83`, released `library.py` SHA-256 `8441de35...e2756e0`, question-generation SHA-256 `1b4cb67f...82d739b`, initial package-bundle SHA-256 `d4c4350a...68b6e7`, and tool-call membership SHA-256 `b73d8637...6f2e17d4`. The supplement packages our controller-audit/certificate code, tests, frozen receipts and matrices, but intentionally does not redistribute the third-party Skill-SP repository; its README gives the exact author commit required for end-to-end rerun. LPs use SciPy HiGHS; certificate classification uses tolerance 10^{-8} . The offline completion analysis is implemented in `asset_first_stri_offline_completion_analysis_20260824.py` and records the frozen Level-1 structured-misspecification families plus deterministic synthetic R^* CPU profiling in a content-addressed JSON artifact. Executable tests cover the target-conditioned dual, target-ray scaling, semantic-first target realization under arbitrary overlap, quotient clone invariance, sampler-weight re-computation, literal-duplicate rejection, exact questioner-message equality, the 7/120 prompt-mixture shift, quotient restoration to zero, one-cell support perturbations, per-tool LP results, and the offline analysis contract. The added E4/E6 analyses use no new support annotations, task outcomes, model calls, or GPU runs. All dynamic thresholds were frozen before outcomes. The failed Qwen3-8B bank remains a transparent non-result; the qualified SkillIRL bridge separately records 18/24 competence calibration, 24 paired A/B/C/D units, exact A/D coupling, and the statistical-resolution boundary that prevents upgrading a realization STOP into a population-level no-effect claim.

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